MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 16 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 5: Arithmetic Progressions** | **MM: 30**

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CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, all intermediate algebraic steps, and clearly boxed final answers.
2. In internal choice (**OR**) questions, both options are solved completely with identical step-marking rigor.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 1.5M, 2M, 3M]**.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason MATHS MANTHAN | (5 × 1 = 5 Marks)

Q1. Which term of the AP $24, 21, 18, 15, \ldots$ is the first negative term? [1]

(A) 8th (B) 9th (C) 10th (D) 11th

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 10th** [0.5 Mark]

Given: $a = 24$, $d = 21 - 24 = -3$.

Formula / Theorem Used: $a_n < 0 \iff a + (n-1)d < 0$

Justification: [0.5 Mark]

$24 - 3(n-1) < 0 \implies 27 - 3n < 0 \implies n > 9 \implies n = \mathbf{10}$ (here $a_9 = 0$, $a_{10} = -3$).

Q2. If the 7th term of an AP is 4 and its common difference is -4, then its first term a is: [1]

(A) 16 (B) 20 (C) 24 (D) 28

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(D) 28** [0.5 Mark]

Given: $a_7 = 4$, $d = -4$.

Formula / Theorem Used: $a_n = a + (n-1)d$

Justification: [0.5 Mark]

$a + 6(-4) = 4 \implies a - 24 = 4 \implies a = \mathbf{28}$.

Q3. In an AP, if $a = 1$, $a_n = 20$, and $S_n = 399$, then n is: [1]

(A) 19 (B) 21 (C) 38 (D) 42

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 1 Mark]

Correct Answer: **(C) 38** [0.5 Mark]

Given: $a = 1$, $a_n = 20$, $S_n = 399$.

Formula / Theorem Used: $S_n = \frac{n}{2}(a + a_n)$

Justification: [0.5 Mark]

$399 = \frac{n}{2}(1 + 20) \implies 798 = 21n \implies n = \frac{798}{21} = \mathbf{38}$.

Q4. The sum of all two-digit odd natural numbers is: [1]

(A) 2475 (B) 2530 (C) 4900 (D) 5050

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(A) 2475** [0.5 Mark]

Given: AP is $11, 13, 15, \ldots, 99$ ($a = 11$, $d = 2$, $l = 99$).

Formula / Theorem Used: $n = \dfrac{l-a}{d} + 1$ and $S_n = \dfrac{n}{2}(a+l)$

Justification: [0.5 Mark]

$n = \frac{99-11}{2} + 1 = 45$; $S_{45} = \frac{45}{2}(11+99) = 45 \times 55 = \mathbf{2475}$.

Q5. **Assertion (A):** If the three interior angles of a triangle are in AP, then one angle must be $60°$.
Reason (R): Taking the angles as $(a-d), a, (a+d)$ gives $(a-d)+a+(a+d) = 180° \implies 3a = 180° \implies a = 60°$. [1]

(A) Both (A) and (R) are true, and (R) is correct explanation.

(B) Both (A) and (R) are true, and (R) is not correct explanation.

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 1 Mark]

Correct Answer: **(A) Both (A) and (R) are true, and (R) explains (A).** [0.5 Mark]

Applying triangle angle sum property:

Formula / Theorem Used: $(a-d)+a+(a+d) = 180° \implies 3a = 180° \implies a = 60°$

Justification: [0.5 Mark]

Reason (R) directly proves why the middle angle of any triangular angle AP must equal $60°$. Both are true and (R) explains (A).

Q6. Find the number of terms in the finite AP $18, 15\frac{1}{2}, 13, \ldots, -47$. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Identifying a, d, and a_n [1 Mark]

$a = 18$, $d = 15\frac{1}{2} - 18 = \frac{31}{2} - \frac{36}{2} = -\frac{5}{2}$, $a_n = -47$.

Formula / Theorem Used: $a_n = a + (n-1)d$

Step 2: Solving for n [1 Mark]

$$18 + (n-1)\left(-\frac{5}{2}\right) = -47 \implies -\frac{5}{2}(n-1) = -65 \implies n - 1 = 65 \times \frac{2}{5} = 26 \implies \boxed{\mathbf{n = 27}}$$

Q7. How many terms of the AP $17, 15, 13, \ldots$ must be taken so that their sum is 72? Explain the double answer. **[2]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Forming Quadratic Equation for n [1 Mark]

Here $a = 17$, $d = 15 - 17 = -2$, $S_n = 72$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

$$\frac{n}{2}[34 - 2(n-1)] = 72 \implies n(18 - n) = 72 \implies n^2 - 18n + 72 = 0$$
$$\implies (n-6)(n-12) = 0 \implies n = 6 \text{ or } 12$$

Step 2: Explaining Double Answer [1 Mark]

Since $a_7 + a_8 + a_9 + a_{10} + a_{11} + a_{12} = 5 + 3 + 1 + (-1) + (-3) + (-5) = 0$, both 6 terms and 12 terms sum to 72.

$\boxed{\mathbf{n = 6 \text{ or } n = 12}}$

Q8. If the nth term of an AP is $a_n = 2n + 1$, find the sum of its first n terms (S_n). **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Identifying a_1 and Last Term a_n [1.5 Marks]

Putting $n = 1$: $a_1 = 2(1) + 1 = 3$. Last term $l = a_n = 2n + 1$.

Formula / Theorem Used: $S_n = \frac{n}{2}(a_1 + a_n)$

Step 2: Simplifying S_n [1.5 Marks]

$$S_n = \frac{n}{2}[3 + (2n + 1)] = \frac{n}{2}(2n + 4) = \frac{n}{2} \cdot 2(n + 2) = \boxed{\mathbf{n(n + 2)}}$$

Q9. The 17th term of an AP exceeds its 10th term by 7. Find the common difference d. **[3]**

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Expressing Both Terms Using $a_n = a + (n - 1)d$ [1.5 Marks]

Let first term $= a$ and common difference $= d$. Then $a_{17} = a + 16d$ and $a_{10} = a + 9d$.

Formula / Theorem Used: $a_{17} - a_{10} = 7$

Step 2: Solving for d [1.5 Marks]

$$(a + 16d) - (a + 9d) = 7 \implies 7d = 7 \implies d = \frac{7}{7} = \boxed{\mathbf{1}}$$

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Q10. If S_1, S_2, S_3 are the sums of the first n terms of three APs each having first term 1 and common differences 1, 2, and 3 respectively, prove that $S_1 + S_3 = 2S_2$. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Given: All three APs share $a = 1$, with $d_1 = 1$, $d_2 = 2$, $d_3 = 3$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$

Step 1: Computing S_1, S_2, and S_3 [3 Marks]

$$S_1 = \frac{n}{2}[2(1) + (n-1)(1)] = \frac{n(n+1)}{2}$$

$$S_2 = \frac{n}{2}[2(1) + (n-1)(2)] = \frac{n}{2}(2n) = n^2$$

$$S_3 = \frac{n}{2}[2(1) + (n-1)(3)] = \frac{n(3n-1)}{2}$$

Step 2: Verifying $S_1 + S_3 = 2S_2$ [2 Marks]

$$S_1 + S_3 = \frac{n(n+1)}{2} + \frac{n(3n-1)}{2} = \frac{n(n+1+3n-1)}{2} = \frac{n(4n)}{2} = 2n^2 = \boxed{\mathbf{2S_2} \quad \text{[Hence Proved]}}$$

Q11. Solve the equation: $2 + 5 + 8 + 11 + \cdots + x = 345$. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Formulating AP Parameters [1 Mark]

Here $a = 2$, $d = 5 - 2 = 3$, last term $a_n = x$, sum $S_n = 345$.

Formula / Theorem Used: $S_n = \frac{n}{2}[2a + (n-1)d]$ and $x = a + (n-1)d$

Step 2: Solving Quadratic Equation for n [2.5 Marks]

$$\frac{n}{2}[2(2) + (n-1)(3)] = 345 \implies n(3n+1) = 690 \implies 3n^2 + n - 690 = 0$$

Splitting middle term ($46 \times (-45) = -2070 = 3(-690)$, $46 - 45 = 1$):

$$3n^2 - 45n + 46n - 690 = 0 \implies 3n(n-15) + 46(n-15) = 0 \implies (n-15)(3n+46) = 0$$

Since $n \in \mathbb{N}$, reject $n = -\frac{46}{3}$; take $n = 15$.

Step 3: Finding $x = a_{15}$ [1.5 Marks]

$$x = 2 + (15-1)(3) = 2 + 14 \times 3 = 2 + 42 = \boxed{\mathbf{44}}$$

Q12. CASE STUDY: City Auditorium Staggered Seating Capacity

A newly inaugurated city auditorium has 25 rows of curved theatre seating. The front row (Row 1) contains 20 seats, Row 2 contains 22 seats, Row 3 contains 24 seats, and each subsequent row holds 2 more seats than the preceding row. [5]

(i) How many seats are installed in the 12th row? [1]

(ii) How many seats are installed in the last (25th) row? [1]

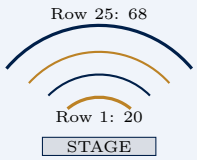
(iii) Calculate the total seating capacity of the auditorium across all 25 rows. [3]

— **OR** —

If every auditorium seat is priced at Rs. 300 for a sold-out evening show, determine the total ticket revenue generated. [3]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Extracted Scenario Data: Row capacities form the AP $20, 22, 24, \ldots$ with $a = 20$, $d = 2$, and $n = 25$.



Formula / Theorem Used: $a_n = a + (n-1)d$ and $S_n = \frac{n}{2}(a + a_n)$

Sub-part (i): Seats in the 12th Row [1 Mark]

$a_{12} = 20 + (12-1)(2) = 20 + 22 = \boxed{\textbf{42} \text{ seats}}$.

Sub-part (ii): Seats in the 25th Row [1 Mark]

$a_{25} = 20 + (25-1)(2) = 20 + 48 = \boxed{\textbf{68} \text{ seats}}$.

Sub-part (iii): Total Seating Capacity Across All 25 Rows [3 Marks]

$$S_{25} = \frac{25}{2}(a_1 + a_{25}) = \frac{25}{2}(20 + 68) = \frac{25}{2} \times 88 = 25 \times 44 = \boxed{\textbf{1,100} \text{ seats}}$$

Sub-part (iii) [OR Option]: Total Ticket Revenue at Rs. 300/Seat [3 Marks]

With total seating capacity $S_{25} = 1{,}100$ seats:

$$\text{Total Revenue} = 1{,}100 \times \text{Rs. } 300 = \boxed{\text{Rs. } \textbf{3,30,000}}$$

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